

International Islamic University Chittagong
Morality Development Program (MDP)
Semester End Examination, Autumn- 2025
Course Title: Tajweedul Quran, Course Code: MDP-1201A,
2nd Semester

Full Marks : 50

Time: 2.5 Hours

[Part [A] Answer each of the following question: 3 x 10 = 30

Answer the following questions

Q. [1] Write down the complete sentence choosing the correct option:

- i. The number of Vowels in Arabic is:
a) Three [3]
b) Five [5]
c) Six [6]
- ii. The letters articulated from the throat are:
a) Three in number
b) Fore in number
c) Six in number
- iii. The term Lahn is interpreted as:
a) Violation of the rules of recitation
b) Beautifying the recitation of the Quran
c) Misinterpretation of the Quran
- iv. Classification of Huruf into voiced and unvoiced is one of:
a) The characteristics that have opposites
b) The characteristics that have no opposites
c) The additional characteristics of Huruf.
- v. Stop is one of the characteristics of Letters:
a) While articulating this the vocal cords vibrate
b) While articulating this the vocal cords do not vibrate
c) While articulating this the airflow is obstructed fully
- vi. The letters combined in the phrase [قطب جد] are:
a) The example of Huruf al-Qalqalah
b) The example of Huruf al-Idhlaq
c) The example of Huruf al Itbag
- vii. The letter [ن] is are of the
a) Letters that is articulated from Cavity of mouth
b) Nasal letters is Arabic
c) Letters that is articulated from the lips.
- viii. The letters Saad [ص], Seen [س] and Zay [ز] are:
a) The examples of Huruf of Safeer (صفير)
b) The examples of Voiced letter (مجهور)
c) The examples of Huruf al-Qalqalah (القلقلة)
- ix. If any of the Huruf al-Halaq (حروف الحلق) is followed by a Noon Sakinah then:
a) It must be recited with Idhhar (إظهار)
b) It must be recited with Idhgham (إدغام)
c) It must be recited with Ikhfaa (إخفاء)
- x. Al-Iqlab (الإقلاب) is transforming:
a) Noon Sakinah or Tanween into the sound of Meem
b) Noon Sakina or Tanween into the clear sound of Noon
c) Noon Sakinah or Tanween is regarded as Ikhfa.

2) 3) Idhhar

Q. [2] Note down the following statements and determine whether they are right or wrong.

- Makharaj al-Huruf means the characteristics of the letters in Arabic *wrong*
- A vowel is that while articulating it the vocal cords vibrate. *right*
- Al-Ikhfa technically refers to a specific rule of pronunciation where the Noon Sakinah is articulated with full clarity. *wrong*
- When Meem Sakinah is followed by another Meem (ميم) then both Meems are merged into other. *right*
- The rule of Ikhfa Shafawi (إخفاء شفوي) is applied in Meem Sakinah in the situation like (ذَلِكُمْ خَيْرٌ لَّكُمْ) *wrong*

OR

Write short note on following terms:

- Al-Isti'adhah and al-Basmalah
- Quranic Mistakes
- Points of articulation
- Definition of Soft letters with examples.



Q. [3] Answer the following questions in a sentence:

- Mention the specific point (Makhrāj) from where the letter [ت] and [د] are articulated.
- What is the situation where Idhhaar Shafawi is applicable for Meem Sakinah? Give an example.
- Mention the situation where the rule of Idhhar (إظهار) is applicable with Noon Sakinah and Tanween. Give example.
- Which of the rules related to Meem Sakinah applicable in the following examples?
يَوْمَ هُمْ بَارِزُونَ - كَمْ مِنْ فِئَةٍ
- Mention which of the following is the example of Madd Muttasil and of Madd Munfasil:

لاَ أُعْبِدُ مَا تَعْبُدُونَ - أُولَئِكَ عَلَى هُدًى مِنْ رَبِّهِمْ

munfasil

PART- [B]: Answer any one of the following questions:

20

Q. [4]: Mention different rules related to Meem Sakinah (الميم الساكنة)

Q. [5]: What do you understand by the term al-Madd al-Fari (المَدَّ الفرعي)? Mention the classification of al-Madd al-Fari with examples.

Wishing you the best of success.

International Islamic University Chittagong
Center for General Education (CGED)
Semester End Examination, Autumn-2025

Course Code: GEED-1201
Full Marks: 50

Course Title: Basic Principles of Islam
Time: 2 Hours 30 Minutes

(Answer all questions. The right side columns contain marks, CLOs, and Bloom's taxonomy domain for each question):

	Questions	Marks	CLOs	Bloom's taxonomy domain
1	a. What are the main objectives of 'Ibadah? Discuss them in the light of Qur'an and Sunnah?	5	3	Remember
	b. Describe some opinions of Muslim scholars regarding the conditions of 'Ibadah.	5		Understand
2	a. Summarize the prerequisites (Shurut) and basic components (Arkan) of Salah.	10	3	Create
	Or,			
	b. Explain the importance of Salah in the light of Qur'an and Sunnah summarizing the consequence of abandoning Salah.	10	3	Create
3	Point out a proper method of Zakah distribution to eradicate poverty from our society.	10	3	Analyze
4	Evaluate the social and religious impact of Sawm (Fasting) in our individual and collective life.	10	3	Evaluate
5	Rate the importance of Hajj and its impact on our life for creating worldwide Muslim Ummah.	10	3	Apply

International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Semester Final Examination, Autumn-2025
Course Code: CSE-1221 Course Title: Computer Programming II
Total marks: 50 Time: 2 hours 30 minutes

[Answer *all* the questions. Figures in the right hand margin indicate full marks.
 Separate answer script must be used for Group A and Group B]

Group A

Q. No.	Question	CLO	BT	Marks
1.a	Define operator overloading. A friend function cannot be used to overload the assignment operator (=). Explain why?	CO1	U	2

1.b	Create a class Complex that represents a complex number with real and imaginary parts (both stored as double values). Overload the following operators for this class: 1. (+) : Complex addition (adding two Complex objects) 2. (=) : Equality comparison (checking if two Complex objects are equal)	CO2	A	4
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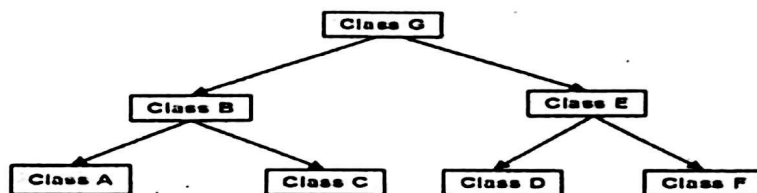
OR

1.b	Create a class Counter that keeps track of an integer count. Overload the ++ operator using a friend function to increase the counter by 1. Include both prefix and postfix versions of the ++ operator. Also, include a function to display the current value of the counter. Write a main() function to demonstrate the increment operations.			
1.c	Write a C++ program to overload both == and != operators for a Vector class using friend functions. The program should compare two vectors and display whether they are equal or not.	CO2	A	4
2.a	Explain how a public data member of a base class behaves when inherited using public inheritance and private inheritance.	CO1	U	2

OR

2.a	How do the properties of the following two derived classes differ? class D1: private B{//....} class D2: public B{//.....}			
-----	--	--	--	--

2.b		CO2	A	4
-----	--	-----	---	---



Consider a hierarchical and multiple inheritance structure involving the classes A, B, C, D, E, and G. Each class contains at least one data member and one member function. Demonstrate how class C can access the information from class B and class G. Similarly, demonstrate how class D can access the information from class E and class G.

OR

2.b	Create a class hierarchy for a bookstore's inventory system with a base class Item (title and price) and two subclasses, Book (author, genre, ISBN) and CD (artist, genre, duration). Subclass further with Fiction and Non-Fiction for books and Rock and Pop for CDs, adding category-specific data. Demonstrate the use of this hierarchy by creating and displaying sample instances of various books and CDs, showcasing their details.			
✓ 2.c	Explain the diamond problem in inheritance and provide a C++ program to demonstrate the solution.	CO2	A	4

CLO Question

CLO BT Marks

Q. No.
 3.a It is permissible for a base pointer to point to a derived object, the reverse is not true." Justify your answer with an example. CO1 U 3

3.b Design a C++ program to manage an employee payroll system using abstract classes. Create an abstract class Employee with the virtual functions calculateSalary() to compute the salary of the employee. Create two derived classes PermanentEmployee, ContractEmployee and implement the pure virtual functions appropriately and display their salary. CO2 C 4

OR

3.b Create a base class named Animal with the following: A pure virtual function void speak() that displays a message indicating the sound made by the animal. Then, create two derived classes: Dog — implements the speak() function to display "Dog says: Woof!". Cat — implements the speak() function to display "Cat says: Meow!". In the main() function: declare an array of pointers to Animal. Dynamically allocate one Dog and one Cat object. Call the speak() function through base class pointers to demonstrate runtime polymorphism. CO1 U 3

3.c Define Binding. Discuss the differences between early binding and late binding with example. CO1 U 3

4.a Describe the key components of C++ exception handling, including try, catch, and throw. CO3 A 3

4.b Suppose you have a program that finds the minimum value between two variables of different data types: integers, floating-point numbers, and characters. The original program uses separate functions for each data type (e.g., Min(int, int), Min(double, double), Min(char, char)). Your task is to re-factor the code using a generic function called Min. The Min function should be capable of finding and returning the minimum of two values of any data type. CO3 A 4

OR

4.b You are given a program that searches for a specific element in an array. The original implementation includes separate search functions for different data types — one for integers, one for floats, and another for characters. Your task is to refactor the program using a generic function named searchElement. The searchElement function should take an array, its size, and a target value, and return whether the value is found in the array regardless of the data type. CO1 U 3

4.c Describe the key components of C++ exception handling including try, catch, and throw with an example. CO1 U 3

5.a What are the differences between Manipulators and ios Functions CO3 U 4

5.b Write a C++ program using formatted I/O operations to show the following output: CO3 A 6

i) ***Advanced C++

ii) ****6.283185

iii) 45.120000***

iv) +7D

v) 000123

OR

5.b Create a user-defined manipulator that formats a floating-point number to be displayed as follows:

i) 12 columns width

ii) Right-justified

iii) Four digits precision

iv) Filling of unused places with zeros

Bismillahir Rahmanir Rahim
International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Semester Final Examination, Autumn 2025
Course Code: CSE-1223 Course Title: Discrete Mathematics
Total Marks: 50 Time: 2 hours 30 minutes

[Answer *all* the questions; in some questions, there are options; answer any one of them; Figures in the right-hand margin indicates full marks. Separate answer script must be used for Group-A and Group-B]

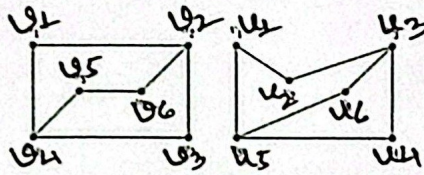
Group-A

- | | | Ma
rks | CLO | DL |
|----|---|-----------|------|----|
| 1. | a) Define <i>linear congruence</i> . Find the solutions of the linear congruence $2x \equiv 3 \pmod{5}$. | 3 | CLO2 | A |
| | b) Explain the <i>encryption</i> and <i>decryption</i> functions of the <i>Caesar cipher</i> . Encrypt the word "HELLO" using the Caesar Cipher with key = 3. (Use A=0, B=1, ..., Z=25)
[OR]
What do you know about <i>pseudorandom numbers</i> and <i>hashing functions</i> ? Find the sequence of three pseudorandom numbers by choosing $m=17$, $a=5$, $c=2$ and $x_0=3$. And If $h(k)=k \pmod{11}$, compute the hash values for keys: $k=23$, $k=45$. | 4 | CLO2 | A |
| | c) Find the value of x using the <i>Chinese remainder theorem</i>
$x \equiv 2 \pmod{3}$
$x \equiv 4 \pmod{7}$
$x \equiv 6 \pmod{10}$ | 3 | CLO2 | A |
| 2. | a) Prove that $\sqrt{2}$ is irrational by giving a <i>proof by contradiction</i> . | 3 | CLO1 | A |
| | b) State the <i>principle of mathematical induction</i> and explain its two main steps. | 3 | CLO1 | U |
| | c) Use mathematical induction to prove that $1 \cdot 1! + 2 \cdot 2! + \dots + n \cdot n! = (n+1)! - 1$ whenever n is a positive integer.
[OR]
Prove that $1^2 - 2^2 + 3^2 - \dots + (-1)^{n-1} n^2 = (-1)^{n-1} n(n+1)/2$ whenever n is a positive integer. | 4 | CLO1 | A |

Group-B

- | | | | | |
|----|---|---|------|---|
| 3. | a) In a robotics lab, each robot is assigned a unique ID tag that is 2 to 4 characters long. <ul style="list-style-type: none"> The first character must be a capital letter (from A to Z). The remaining characters (if any) can be either capital letters or digits. However, due to system constraints, 20 specific ID tags are already reserved and cannot be assigned. How many valid robot ID tags can be created under these rules?
[OR]
A company produces custom license plates that are made up of 7 characters using the letters in the word GARAGES. How many distinct license plates can be created using all the letters of GARAGES? | 4 | CLO2 | A |
| | b) State the <i>pigeonhole principle</i> . In a class of 30 students, at least how many students will have birthdays on the same day of the week? | 3 | CLO2 | A |
| | c) State <i>Inclusion-Exclusion principle</i> . A digital lock generates 8-digit access codes made up only of 0s and 1s (bit strings). How many different access codes are possible that either start with 11 or end with 00? | 3 | CLO2 | A |

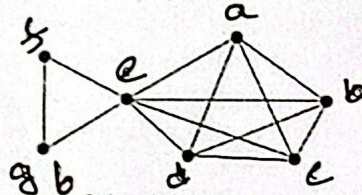
4. a) State the *handshaking theorem*. Are the following two graphs *isomorphic*? Justify your answer. 3 CLO3 /



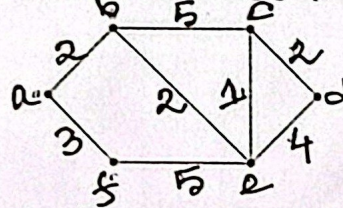
- b) Define *Euler circuit* and *Euler path*. Determine whether the following graph has an Euler circuit or Euler path. Construct such a circuit or path if it exists. 3 CLO3 /

[OR]

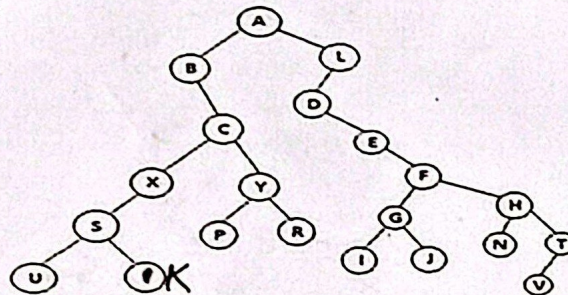
Define *Hamiltonian circuit* and *Hamiltonian path*. Determine whether the following graph has a Hamiltonian circuit or Hamiltonian path. Construct such a circuit or path if it exists.



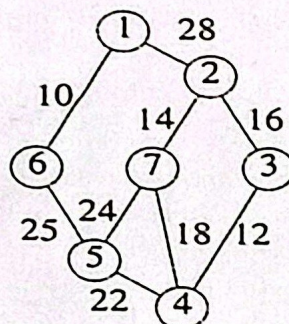
- c) Find the *shortest path* from a to e of the following graph using *Dijkstra's algorithm* 4 CLO3 /



5. a) Define a *tree*. Differentiate between a *graph* and a *tree*. 2 CLO3 /
b) Find the *inorder*, *preorder*, and *postorder* traversal of the following tree. 3 CLO3 /



- c) Define *minimum spanning tree (MST)*. Find the minimum spanning tree of the following graph by using *Kruskal's algorithm*. 5 CLO3 /



[OR]

Define *minimum spanning tree (MST)*. Find the minimum spanning tree of the above graph by using *Prim's algorithm*.

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Final Exam, Autumn-2025

Course Code: **EEE-1221**

Time: 2 hours 30 minutes

Course Title: **Electronics**

Full Marks: 50

The figures in the right-hand margin indicate full marks

Part A

[Answer the questions from the followings]

1. a) What is JFET? Draw the symbol of N-Channel JFET and P-Channel JFET. CO2 U 5
Describe the working principle of N-Channel JFET.
OR,
What is Pinch off voltage in MOSFETs? Describe the basic working principle of an enhancement-mode N-channel MOSFET.

1. b) Sketch the input and output characteristics of an n-channel enhancement type MOSFET, where $V_T = 2\text{ V}$ and $k = 0.5 \times 10^{-3} \text{ A/V}^2$. CO3 A 5

2. a) Can a transistor be used both as a switch and an amplifier? If so, explain its switching action with necessary illustrations. CO2 U 5

OR,

Explain what is multivibrator in electronics. Using a diagram, describe the operational principles of a multivibrator that generates a square wave output.

2. b) Fig. 2(b) shows the transistor switching circuit. Given that $R_B = 2.7\text{ k}\Omega$, $V_{BB} = 2\text{ V}$, $V_{BE} = 0.7\text{ V}$ and $V_{knee} = 0.7\text{ V}$. CO3 A 5

i) Calculate the minimum value of β for saturation.

ii) If V_{BB} is changed to 1V and transistor has minimum $\beta = 50$, will the transistor be saturated.

Draw the output voltage waveform of the circuit given in Fig. 2(b).

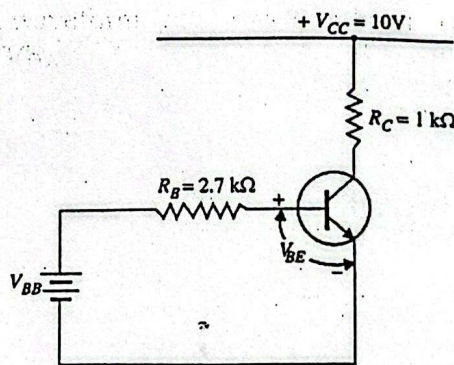


Fig. 2(b)

Part B

[Answer the questions from the followings]

3. a) Sketch a neat diagram and derive an expression for an OP-AMP to use it as a summing amplifier. CO2 U 5

3. b) Draw the output voltage waveform with proper mathematical expression for the circuit given in Fig. 3(b). CO3 A 5

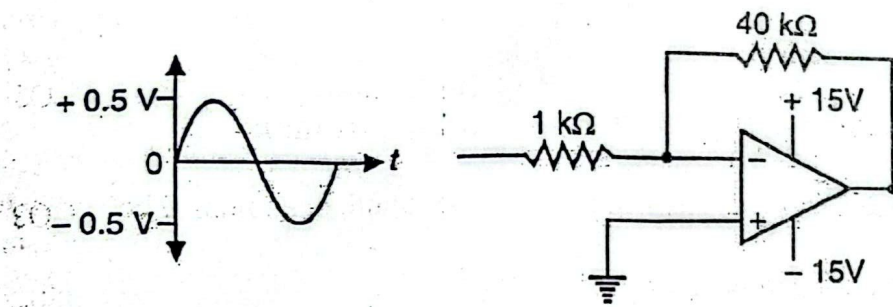


Fig. 3(b)

4. a) What is negative feedback? Derive the relation between open-loop gain and overall gain after feedback.

OR

Show that the input impedance of amplifier increases due to negative voltage feedback.

4. b) The gain of an amplifier without feedback is 50 whereas with negative voltage feedback, it falls to 25. If due to ageing, the amplifier gain falls to 40, Find the percentage reduction in stage gain

(i) without feedback and (ii) with negative feedback

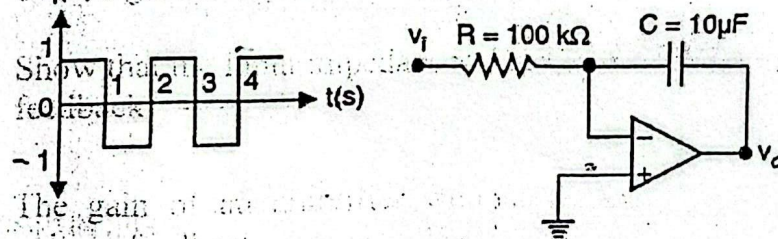
5. a) What is Oscillator? Briefly explain damped and undamped oscillations with proper illustrations.

OR

Design and explain a circuit that can detect the peak value of an incoming unknown varying signal.

5. b) Analyze the output voltage in the context of Figure 5b, where an op-amp integrator is connected to a square wave input.

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International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE
Final Exam, Autumn-2025

Course Code: **PHY 1201**
Time: 2 hours 30 minutes
Marks: 50

Course Title: **Physics-II**
Full

The figures in the right-hand margin indicate full marks

Part A

[Answer the questions from the followings]

1. a) Define crystal systems. Write the seven crystal systems with example. CLO1 R 3
1. b) Show that in a crystal of cubic structure, the distance between the planes with Miller indices h, k, l is equal to $d = \frac{a}{\sqrt{h^2+k^2+l^2}}$, where a is the lattice parameter. CLO1 U 5

Or,

Define coordination number. Explain coordination number for NaCl structure.

1. c) Lead is face-centered cubic with an atomic radius of $r = 1.674$ A.U. Find the spacing of (220) planes. CLO2 A 2

2. a) How many types of crystal defects in solid? Explain all of the crystal defects. CLO1 U 5

Or,

○ Distinction between metal, insulator and semiconductor in terms of energy band.

2. b) Show that the packing fraction for an fcc lattice is ~ 0.74 and hence write the meaning of this number. CLO2 A 5

Part B

[Answer the questions from the followings]

3. a) What is the meaning of mass-energy relation? CLO1 U 2
3. b) Derive Einstein's mass-energy relation. CLO1 A 5
3. c) The particle of mass 10^{-24} kg is moving with a speed of 1.8×10^8 m/s. Calculate its mass when it is motion. CLO3 An 3

Or,

The length of a rocket on the ground is 100 m. When rocket moves with respect to a stationary observer on the ground its length appears to be 99.5m. Calculate the speed of the rocket?

5. a) (The answer is 70 days)
4. a) Calculate the following properties of ${}^6\text{C}^{12}$ particle: CLO2 A 7
- | | |
|--|----------------------------|
| i. Nuclear Mass | ii. Nuclear Size |
| iii. Nuclear Density | iv. Nuclear Charge |
| v. Nuclear Mass defect | vi. Nuclear Binding Energy |
| vii. Nuclear Binding Energy per nucleon. | |

Or,

Discuss Bohr Atom Model and hence deduce the expression for the calculation of the radii.

$$r = n^2 a_0$$

4. b) Calculate the threshold frequency and corresponding wavelength of radiation incident on a certain metal whose work function is $3.31 \times 10^{-19} \text{ J}$. CLO3 An 3
 Given that Plank's constant = $6.61 \times 10^{-34} \text{ J-s}$? $\frac{1}{2} \text{ m.v.2}$
5. a) Explain the binding energy and nuclear fusion. CLO1 U 2
5. b) Justify the statement "the number of radioactive atoms decreases exponentially with time". CLO2 Ap 5
5. c) The half-life of radon is 70 days, calculate decay constant and mean life of radon? CLO3 An 3